Final Project: Pick a Project

Computer Science - Week 10

Oct 3, 2026 - Version 0.0.2

Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Bohdan Hartsky	Bohdanh@udel.edu
Friend (optional)		

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

I am trying to create an easy-to-use unit converter for chemical engineers.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

Google search, Unit Converter, Pearson.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

I want this web application to be used in everyday chemical calculation.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

Unit Conversion Hub. The goal for the website is to be able to convert different units of measurement into other units of measurement through functions and separate pages for the different types of measurements.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

I would need to provide the user with buttons for what type of measurement they are doing.

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

It will identify the measurement variable being used (m, L, yd), and they will be able to select what they want the measurement converted into.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

They will get back the calculated unit they input converted into the unit they selected.

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

What it tracks is the unit and text input of the user such as cm and volume.

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

1. *Integer input* 2. *Unit selection* 3. *Conversion unit selection* *3.*

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

The most difficult would be converting from larger units of measure to another.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

Monday, Wednesday and friday and maybe during the weekends

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

My ultimate goal for this project is to build something I can use and expand upon for use in other classes.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

What are common errors students make with unit conversion?

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

Common unit-conversion mistakes include:

- **Using the conversion factor backward:** For example, treating $1m = 100cm$ as $1cm = 100m$.

Best habit: Write the conversion as a fraction so the unwanted unit cancels out:

$$5m \times \frac{100cm}{1m} = 500cm$$

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

Instead of conversion in the unit I would like to focus more on the conversions of the base units for now (Ex: m, J)

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

I want to focus on the base units so the idea is more focused on converting the units without making it too complex such as converting m into cm.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

I can use the LLM provided to help with any errors. Also, I can use it to ask for example code.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

I would look at the error code and the trace to see where exactly it went wrong and make the changes based on the line that is causing the error.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

ChatGPT:

What are common errors students make with unit conversion?

Common unit-conversion mistakes include:

- ***Using the conversion factor backward:*** *For example, treating* $1m = 100cm$ *as* $1cm = 100m$.
- ***Forgetting to convert all units:*** *In a problem involving meters and seconds, students may convert meters but leave seconds unchanged.*

- ***Multiplying when they should divide (or vice versa):*** *Dimensional analysis helps catch this.*
- ***Dropping or adding zeros incorrectly:*** *Especially with metric prefixes like milli-, centi-, and kilo-.*
- ***Confusing squared or cubed units:*** $1m^2 = 10,000cm^2$*, not* $100cm^2$*.*
- ***Not checking the direction of the answer:*** *Converting meters to centimeters should produce a* ***larger numerical value****, since centimeters are smaller units.*
- ***Rounding too early:*** *Keeping extra digits until the final step prevents accumulated rounding errors.*
- ***Mixing up different systems:*** *For example, confusing miles with kilometers or pounds with kilograms.*

Best habit: *Write the conversion as a fraction so the unwanted unit cancels out:*

$$5m \times \frac{100cm}{1m} = 500cm$$

The canceled unit provides a quick check that the setup is correct.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.

Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have

not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

Loops will help me have the number go through if statements and if it is true the required conversion with occur.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

The easiest part will be the drafter because of how small it is in comparison to the other code in the project. The hardest will be the implementation of the functions.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

I should implement the different functions first amd test them out because they are going to be the hardest part of the project.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Distractions such as other homework and studying will interfere, but I will manage my time.

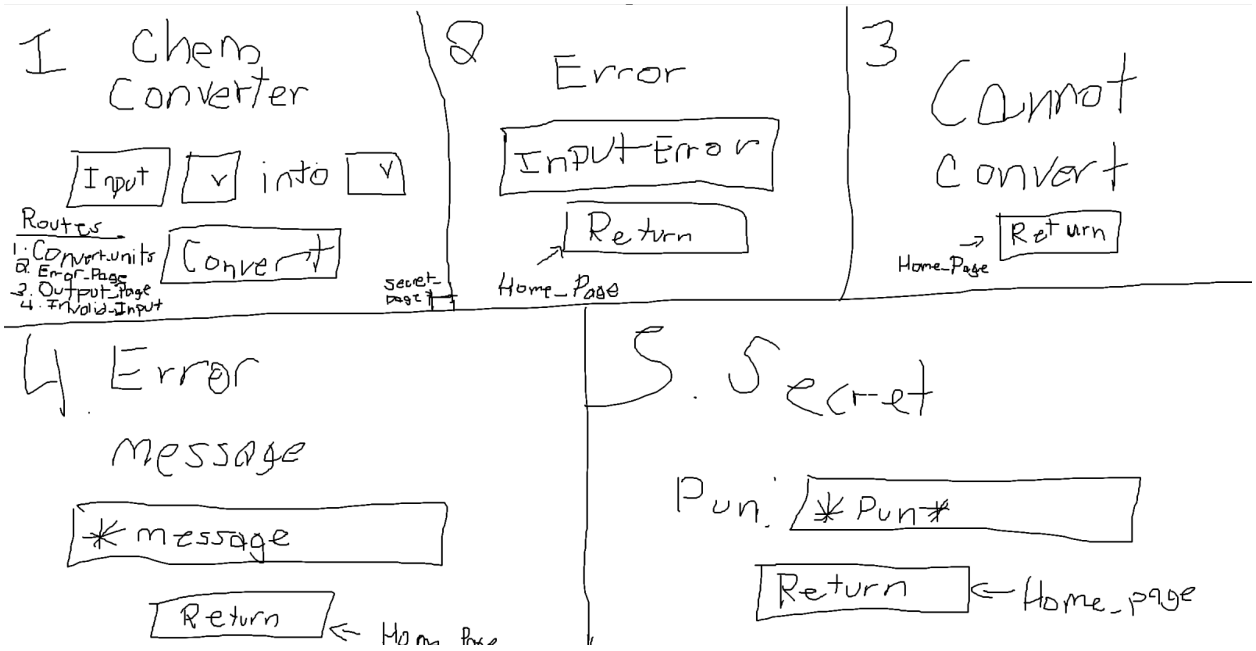
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.


1 Chem Converter
Input
v
into
v
Routes
1. Convert_units
2. Error_Page
3. Output_Page
4. Invalid_Input
Convert
Secret_Page
2 Error
Input Error
Return
Home_Page
3 Cannot Convert
Home_Page
Return
4. Error message
*message
Return
Home_Page
5. Secret
Pun
Pun
Return
Home_page

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	**The purpose of my website is to be able to convert an input of a selected unit from the user into another which the user will also select.**
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	State: - UserInput: float - Unit: list[Symbol] - Output: float - Convertible: bool
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	Symbol: Symbol: int Unit: str
Routes (minimum 7) The names of your route functions	Routes: - Home_page - Convert_units - output _page - Error_page (error message) - Invalid_input_page

	- Cannot_convert_page - Secret_page
Pages (minimum 5) A brief description of the pages in your site	- Main Page: Takes in user input and sends them to another page with the output with a small button in the corner (well hidden) that takes you to Secret Page. - Output (1): The output was able to occur correctly and it gives it's output. - Output (2): Error output with a button back to the homepage or to the page that lists the error. - Output (3): Error page for "cannot convert" - Output (4): Lists the error that caused it. - Secret Page: An Easter egg someone can find that generates a chemistry pun.
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	TextBox - Allows user to enter the number they want to convert SelectBox - Allows user to input the unit they want to convert SelectBox - Allows use to input the unit they want to convert into Button - A pressable button that triggers the conversion and takes the user to a separate page.
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	- If the unit == user input unit (do this for every unit I want to do) Convert unit into output unit - If error = "": - Display error message based upon what the error is
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in	The for loop will iterate through state.units, find the unit, and select a conversion function based on that unit.

your State attribute. Description of this loop and why it is meaningful to the site	

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

Replace this text with your answer.

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.